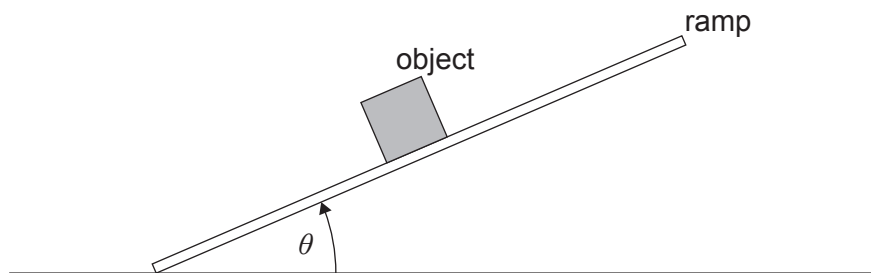


PAPER 1 / QUESTION 53

2 May 2023 / TZ2 (file label) / HL Paper 1 / original question 5 / 2223-6513

53. The angle θ between a ramp and a horizontal surface slowly increases from zero. An object on the ramp does not slide as θ increases.



The frictional force on the object is

- A. constant.
- B. negligible.
- C. proportional to $\cos \theta$.
- D. proportional to $\sin \theta$.